CM0131

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

MARKING SCHEME

JUNE 2013

CHEMISTRY 9189/4

A/W changed, deformation.

1 (a) Disruption of bonds in tertiary/secondary structure of protein; [1] [1]

by extremes of – (high) temperature – bonds broken [1] [(H) bonds] [1]

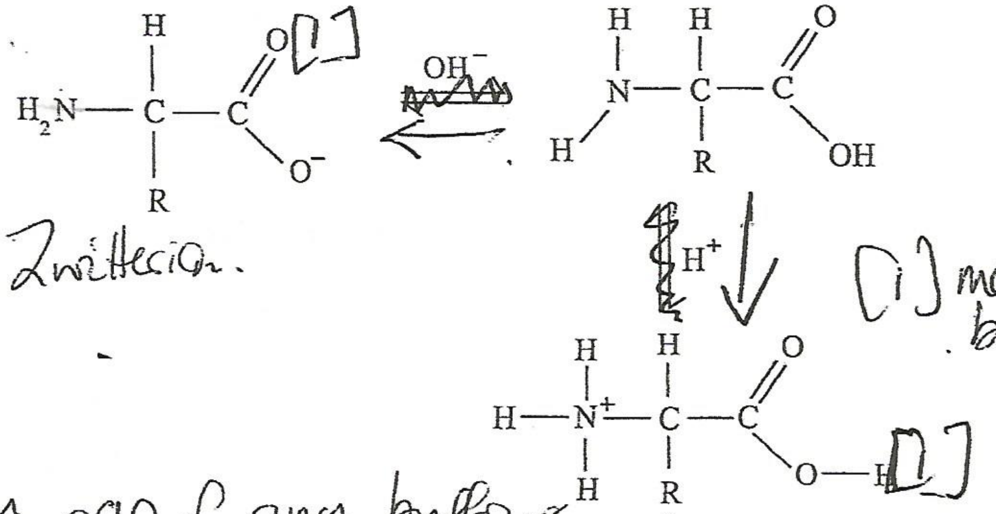
/ pH – ionic bonds protonated or deprotonated

can form hydrogen bonds; [1] / Presence of heavy metal ions.

subrate mark the any ms

(A) relevant diagram showing denaturation

(b) Buffer activity – ability to resist small changes in pH; [1]

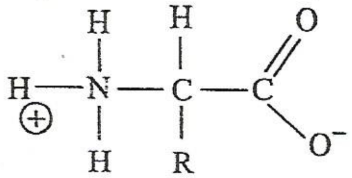


(A) Zwitterion.

[1] mark each both equation

(A) any eqn of any buffering system e.g. HCO_3 in blood. [1]

(c) (iii) (Zwitterion – dipolar amino) acid. / AW.



/ diagram.

[1]

(d) (iv) Isoelectric point – pH at which amino acid exists (as a zwitterion) with no (overall) charge [1]

(e) (v) Reversible – inhibitor ~~similar to substrate and can fit into active site~~ binds reversibly [1] to the active site [1] enzyme.
[substrate determines rate; [1]
~~Km , Vmax reached~~; [1]

irreversible – inhibitor binds ~~another site and alters active~~ site; irreversibly [1] to the enzyme [1] alter the shape [1]
substrate can no longer fit
Km and Vmax altered; / A.W. blocking substrate [1]

2 (a) Temperature; decrease inactivates; enzyme activity / refrigeration [2]

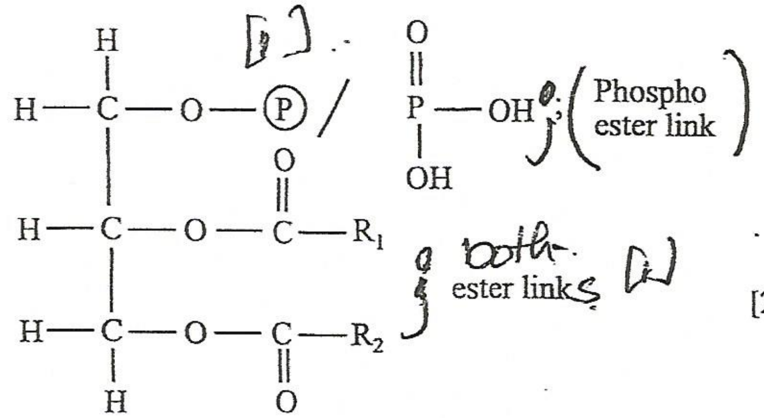
pH; increase in acidity / alkalinity stops enzyme action; [2]

(b) (i) [1] 3 Dimensional folding; control / diagram. [1]
held by VDW/Ionic /Disulphide/Hydrogen bonds [at least 2] [1] any two [2]

(ii) Carves out active site; [1]
which enables substrate to produce a perfect fit; [1]
bringing reacting groups together; [1]

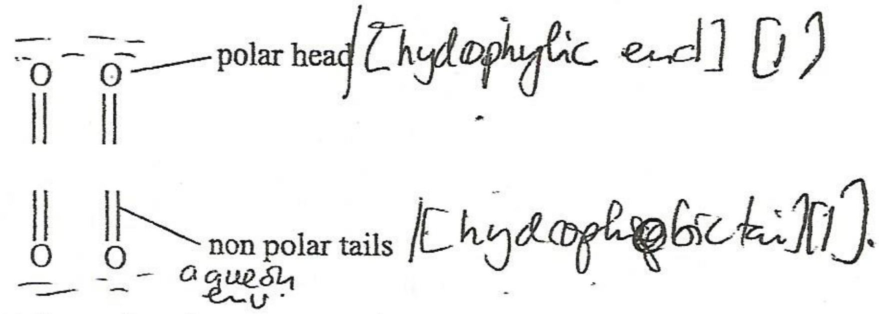
[Total: 10 marks]

3 (a) (i)



[2]

(ii)



Align to shield themselves from aqueous / non aqueous environment [2]

(b) (i) movement of molecules across membrane against their concentration gradient with use ATP / energy; / ans. [1]

(ii) channel proteins aid polar molecules to move through non polar region;
(movement) of Na^+ for K^+ is to maintain ~~electrical balance;~~ osmotic pressure / transmission of nerve impulse [1]
/ protein biosynthesis.

(iii) inclusion of large molecular weight proteins which increases bond strength; weaker bonds [van der Waals] non polar tails. [2] other

(iv) Any function [i.e hormonal receptors cell to cell recognition]; [1]

[Total: 10 marks]

4 (a) (i) - concentration of pollutants rise during early morning rush hour; [1]

4 AW

- NO_2 builds up as NO is oxidised; [1]

Presence of soot smoke, dust at high concentration.

Any 4 pts.

- $[NO_2]$ falls due to photolysis, which leads to formation of ozone and aldehydes;

Around noon ozone, aldehydes increase, due to photolysis of NO_2 producing oxygen radicals.

- NO_2 and hydrocarbon produced in the afternoon rush hour are removed by reaction with ozone leading to drop in ozone; [1]

(ii) - sunlight; [1]

- temperature inversion ; [1]

any two pts.

- still air and dust particles; [1]

- Presence of many primary pollutants.

max 2

(b) (i) $SO_{2(g)} + NO_{2(g)} \rightarrow SO_{3(g)} + NO_{(g)}$; [1]

$SO_{3(g)} + H_2O_{(l)} \rightarrow H_2SO_{4(aq)}$ [1]

(ii) $NO_2 \xrightarrow{hv} NO\bullet + O\bullet$; [1]

$O^{*} + O_2 + M \rightarrow O_3 + M^{*}$; [1]

[Total:10 marks]

5 (a) - Food industry.

- as a solvent; [1]

- transport.

- for cooling; [1]

Any 3

- bottled water.

- washing away waste; [1]

- Irrigation.

(b) (i) Adding of chlorine to water / to kill bacteria; / AW. [1]

(ii) - chlorine kills harmful bacteria; [1]

- production of chlorinated organic matter which can be carcinogenic; / dioxins. [1]

(c) (i) - it kills harmful bacteria; / [1] [1]

- remove taste and odours in water;

(ii) - not easily available; / [1]

- cannot protect water after it leaves the treatment plant; [1]

(iii) - presence of Ca^{2+}; and [1]

Mg^{2+} ions; [1]

[Total:10 marks]

(R) one without equilibrium sign.

6 (a) (i) $O_{2(g)} + 4H^{+}_{(aq)} + 4e^{-} \rightleftharpoons 2H_2O_{(l)}$ + 1.23 V; [1]

(ii) - conditions within soils are not standard; [1]

- concentration of oxygen and also the acidity are low; [1]

(iii) water logged soils / Padi fields [1]

$Fe^{3+}_{aq} + e^{-} \rightleftharpoons Fe^{2+}_{aq}$; [1]

$NO^{-}_{3(aq)} + 10H^{+} + 8e \rightleftharpoons NH^{+}_{4(aq)} + 3H_2O_{(l)}$; [1]

$SO_4^{2-}aq + 8H^{+}_{aq} + 8e \rightleftharpoons S^{2-}_{aq} + 4H_2O$; [1]

(b) O — C — O
→ ← → asymmetrical; [1]

O — C — O
← → symmetrical; [1]

↑
O — C — O bending; [1]
↓ ↓

[Total: 10 marks]

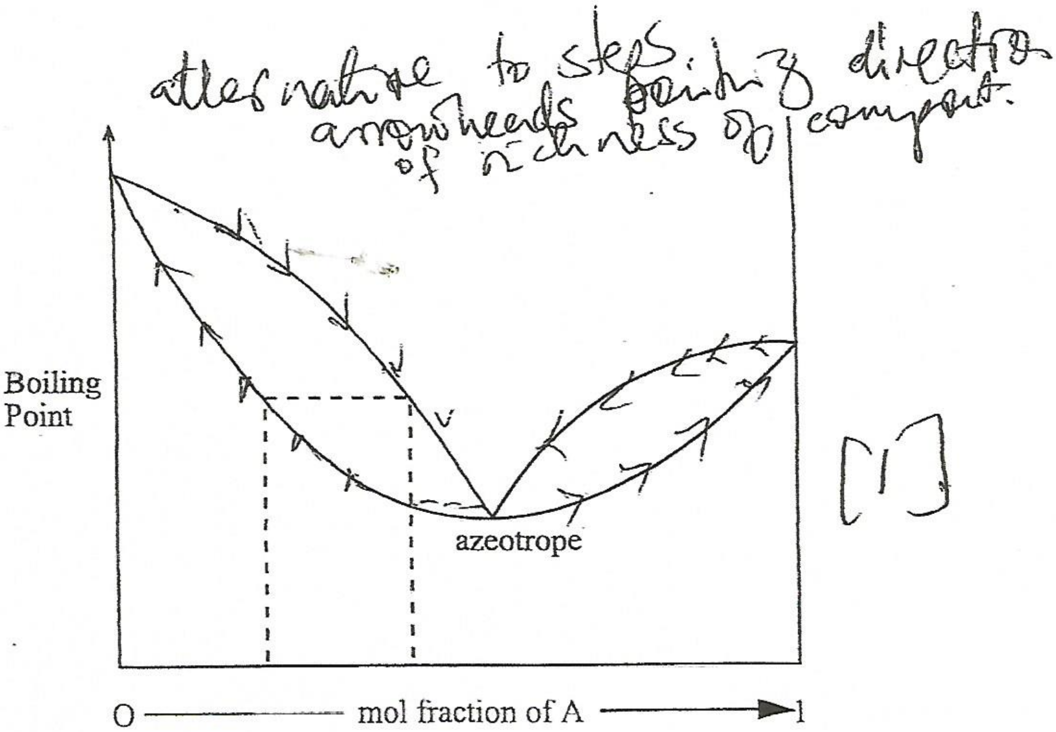
7 (a) (i) composition of a mixture which melts at a ~~lower~~ and constant temperature; [1]

Homogeneous;
(ii) metallic solid mixture containing 2 or more metals; / (A) metal + non-metal [1]

(iii) mixture of liquids with a constant boiling point; [1]

(b) improved hardening; [1]
improved corrosion / tarnishing resistance; [1]
low melting point; [1]
improving tensile strength.
improving appearance.

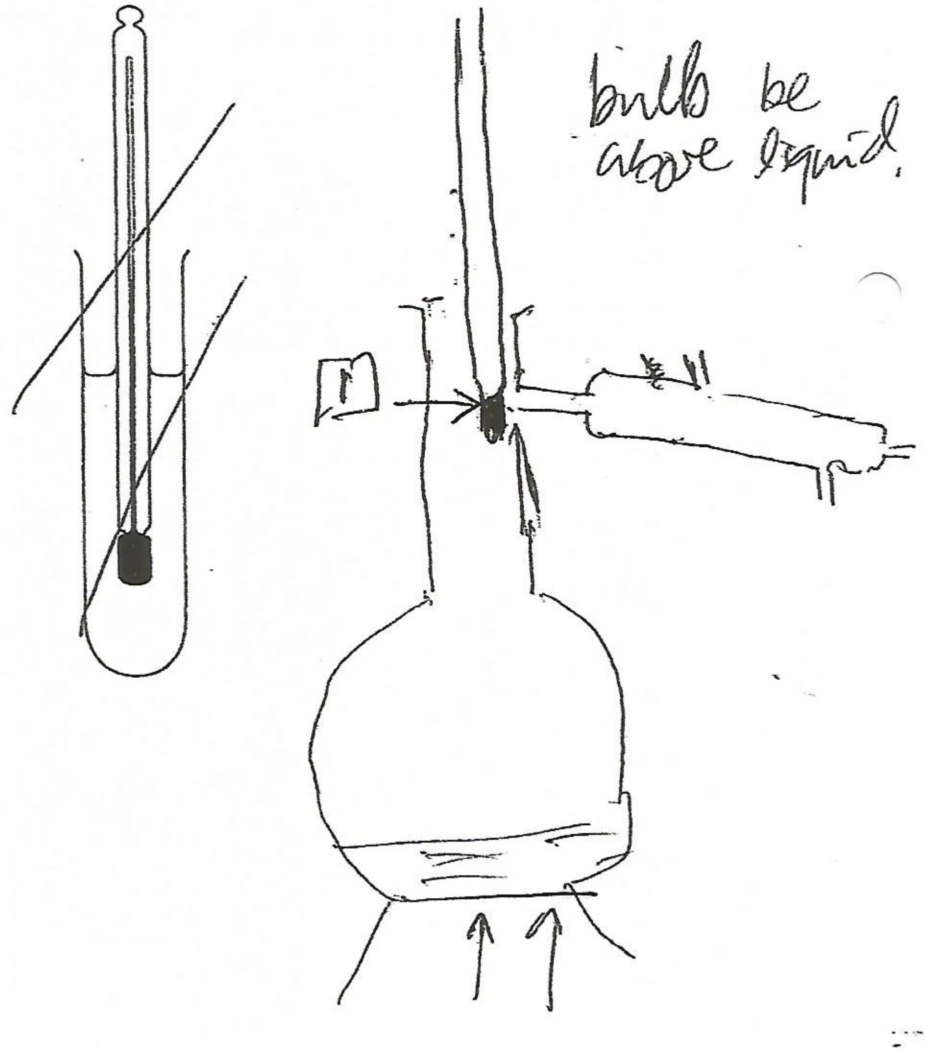
(c) cannot be fully separated by fractional distillation; [1]



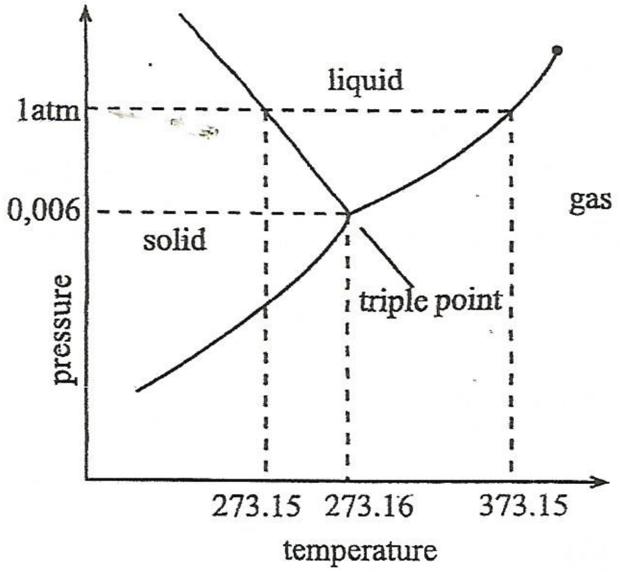
(i) diagram showing areotrope and correct axis; [1]

demonstrating that pure substance cannot be obtained on the diagram; [1]

(ii)



8 (a)



(i) correct shape; [1]

three regions correctly labelled; [1]

axis and boiling,freezing point shown correctly; [1]

triple point labelled; Correct temp^t pressure p^t [1]

(b) (i) negative gradient; [1]

liquid water expands on freezing; [1]

due to increased number of hydrogen bonds / open tetrahedral [1]

(ii) ΔH solution for potassium nitrate more endothermic; [1]

(c) identification of drugs/doping substances analysis of pharmaceuticals; [1]

determining nature/amount of insecticides/pesticides; amino acids [1] } any two [1]
after protein hydrolysis

[Total:10 marks]

9 (a) (i) solubility of a gas is directly proportional to its partial pressure above the solution/AW; (A) mathematical e [1]

(ii) before opening
soft drinks sealed with carbon dioxide at high pressure; [1]

more carbon dioxide dissolved at high pressure; (A) Le Chateliers p [1]
$CO_2(g) \rightleftharpoons CO_2$

pressure drops to atmospheric pressure on opening; [1]

Before opening equilibrium lies to the right.
After opening equilibrium lies to the left.

Ⓐ ~~mathematical expression~~

(b) solubility = Henry's constant × pressure of nitrogen / $S = k_H P_{nitrogen}$ [1]

$S = 7 \times 10^{-4}\ mol\ dm^{-3} atm^{-1} \times 0.78 atm$

$= \quad 5.46 \times 10^{-4} mol\ dm^{-3}$; [1]

(c) ammonia reacts with water / Forms H-bonds with water; [1] NH_3 highly soluble; [1]

$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$; Ⓐ diagram. [1]

(d) separating oils from fowers / [1]

~~separating complex ores~~; [1]

max 1

[Total:10 marks]

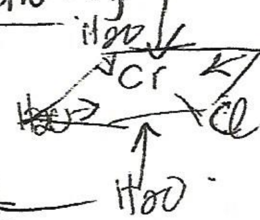
10 (a) (i) Cr – ~~[Ar]~~ $1s^2 2s^2 2p^6 3s^2 3p^6$ $3d^5$ $4s^1$; [1]

[1] - dative
[1] - shape
[1] - charge

(ii) ~~+7~~; ~~blue~~; +6; yellow; [2]

(iii) correct shape; octahedral [2]

cis / trans.



1 mark charge, 1 - shape diagram with dative bond.

(b) ~~The yellow solution turns orange;~~ [1]

~~$2CrO_{4\,(aq)}^{2-} + 2H^+_{(aq)} \rightleftharpoons Cr_2O_{7(aq)}^{2-} + H_2O_{(l)}$~~ [1]

(c) $Cr_2O_{7\ aq}^{2-} + 14H^+_{aq} + 6e \rightleftharpoons 2Cr^{3+} + 7H_2O + 1.33V$; [1].

EC-f ??

$H_2O_{2(aq)} + 2H^+_{(aq)} + 2e^- \rightleftharpoons 2H_2O_{(l)} + 1.77V$; [1]

Ⓡ without units but do not penalise twice.

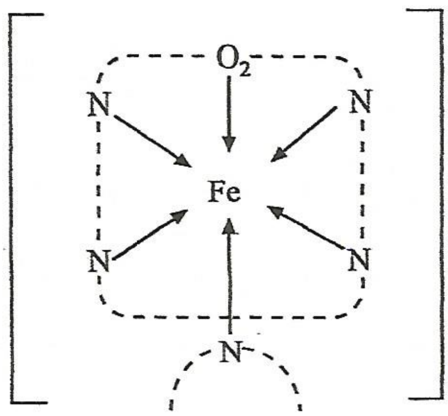
$3H_2O_2 + 2Cr^{3+} + H_2O \rightarrow Cr_2O_7^{2-} + 8H^+$; [1]

$E^\theta cell = 0.44\ V$; [1]

[Total:10 marks]

11 (a) (i) Pale ~~[illegible]~~ Green; Ⓡ green alone. [1]

(ii)



(iii) Carbon monoxide binds to haemoglobin irreversibly at the site of oxygen; [1]
Oxygen can no longer be transported; [1]

(iv) - using pure oxygen; increases conc of O_2 [1]
concentration of oxygen is very high; [1]

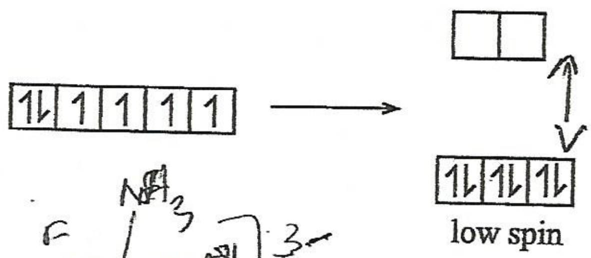
(b)

K	Fe	O	
$\frac{39.4}{39}$	$\frac{28.3}{55.8}$	$\frac{32.3}{16}$;	[1]
$\frac{1}{0.5}$	$\frac{0.5}{0.5}$	$\frac{2}{0.5}$;	[1]
	K_2FeO_4;	+6;	[1]

ecf on ox

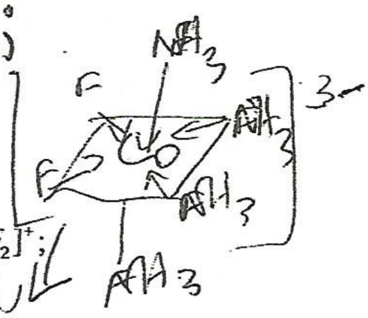
12 (a) (i)

correct splitting;
electron pairing;



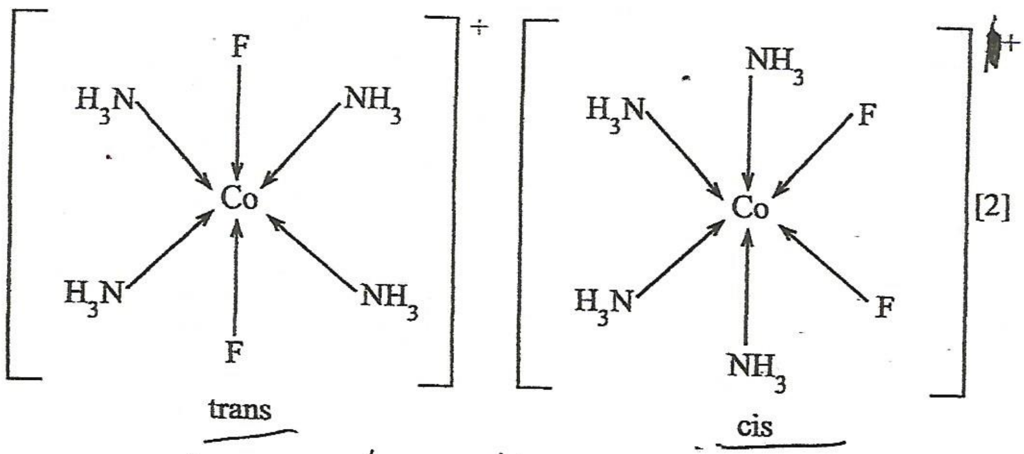
[2]

(ii) Octahedral; [1]



(iii) $[CO(NH_3)_4F_2]^+$; [1]

(iv)



[2]

(b) Alloys; pigments; paints; vitamin B12 [1]

(c) ~~$CO^{3+} + e^- \rightleftharpoons CO^{2+}$ +1.82 V~~ $Co^{3+} + e^- \rightleftharpoons Co^{2+}$ +1.82

$O_2 + 4H^+ + 4e^- \rightleftharpoons 2H_2O$ +1.23 V; [1]

~~$HCO^{3+} + 2H_2O \rightarrow 4CO^{3+} + H_2O$;~~

$4Co^{3+} + 2H_2O \rightarrow 4Co^{2+} + 4H^+ + O_2$ [1]

$E^\theta cell = 0.59$ V; [1]

[Total:10 marks]